A Complete Critical Holder Endpoint Map for RKHS Sandwiches

Agent lane 7

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Source and Target

The source is Max Schölpple and Ingo Steinwart, *Which Spaces can be Embedded in Reproducing Kernel Hilbert Spaces?*, arXiv:2312.14711 [1]. The paper proves that, for Banach function spaces $E \hookrightarrow F$, an intermediate reproducing kernel Hilbert space $E \hookrightarrow H \hookrightarrow F$ exists if and only if the inclusion $E \hookrightarrow F$ is 2-factorable. It also records the necessary consequence that such an inclusion must be of cotype 2.

For bounded $X \subset \mathbb{R}^d$ with nonempty interior, the source proves that no RKHS sandwich

$$\mathrm{Hol}^\alpha(X) \hookrightarrow H \hookrightarrow \mathrm{Hol}^\beta(X)$$

can exist when $\alpha - \beta < d/2$. In the Besov/Triebel-Lizorkin section, it proves positive RKHS sandwiches when the smoothness gap is strictly larger than $d/2$, and identifies the critical line $\alpha - \beta = d/2$ as a border case.

these preparations our result reads as follows.

Theorem 12. *Let (X, d) be a metric space and $0 < \beta \le \alpha \le 1$. If X is connected and there exists an RKHS H over X with*

$$\mathrm{H\ddot{o}l}_d^\alpha(X) \hookrightarrow H \hookrightarrow \mathrm{H\ddot{o}l}_d^\beta(X)\,, \tag{12}$$

then there exists a constant $C > 0$ such that for all $0 < \delta < \min\{1, \operatorname{diam} X\}$ we have

$$N_d^{\mathrm{pack}}(X, \delta) \le C\delta^{-2(\alpha-\beta)}\,. \tag{13}$$

Moreover, if there exists an RKHS H over X such that

$$\mathrm{H\ddot{o}l}_d^\alpha(X) \hookrightarrow H \hookrightarrow \ell_\infty(X)\,, \tag{14}$$

then there exists a constant $C > 0$, such that (13) *holds for $\beta := 0$ and all $0 < \delta < \min\{1, \operatorname{diam} X\}$.*

Let us consider a bounded $X \subset \mathbb{R}^d$ with non-empty interior. By Theorem 12 and the packing number bound (8) we conclude that there is no RKHS H satisfying (12) if $\alpha - \beta < d/2$. Moreover, there is no RKHS H satisfying (14) if $\alpha < d/2$. Hence, (1) is infeasible for $d > 2$ within the family of Hölder spaces. We observe that the difference of the smoothness between the Hölder spaces in (12) needs to grow proportional to the underlying dimension in order to enable (1).

In Section 4.5, (26) will generalize Hölder spaces over X, allowing arbitrarily large non-integer smoothness. In Theorem 21 we will show that if $\alpha - \beta > d/2$, then there exists an intermediate RKHS H such that we have

$$\mathrm{H\ddot{o}l}^\alpha(X) \hookrightarrow H \hookrightarrow \mathrm{H\ddot{o}l}^\beta(X)$$

and in Theorem 22 we will show that for $\alpha > d/2$ there exists an intermediate RKHS H such that

$$\mathrm{H\ddot{o}l}^\alpha(X) \hookrightarrow H \hookrightarrow C^0(X)$$

holds. Hence, the bounds derived in Theorem 12 are tight up to the border cases $\alpha - \beta = d/2$, respectively $\alpha = d/2$. For general metric spaces (X, d) however, we do not know any condition that implies the existence of an RKHS H satisfying (12) or (14).

The requirement that X is a *bounded* subset of $\mathbb{R}^d$ is crucial to obtain positive answers to (1), as the following

Critical Border Map

Write
$$\Lambda^s([0,1]^d) = B^s_{\infty,\infty}([0,1]^d)$$
for the Holder-Zygmund space. For noninteger $s = n + \theta$, $n \in \mathbb{N}_0$, $0 < \theta < 1$, this is the usual classical Holder space $C^{n,\theta}$, up to equivalence of norms. For integer s, Λ^s is the Zygmund endpoint, not the smaller classical space C^s and not $C^{s-1,1}$.

Theorem 1 (Holder-Zygmund critical line)**.** *Let $d \geq 1$, $\beta > 0$, and*
$$\alpha = \beta + d/2.$$
Then the inclusion
$$\Lambda^\alpha([0,1]^d) \hookrightarrow \Lambda^\beta([0,1]^d)$$
is not cotype 2. Consequently, there is no reproducing kernel Hilbert space H on $[0,1]^d$ such that
$$\Lambda^\alpha([0,1]^d) \hookrightarrow H \hookrightarrow \Lambda^\beta([0,1]^d).$$

This settles the critical Holder-Zygmund/Besov $B^s_{\infty,\infty}$ line negatively, including integer values of α. The integer endpoint is convention-sensitive, however:

Proposition 1 (Classical integer domain: positive Zygmund target)**.** *Let $m \in \mathbb{N}$, $d < 2m$, and put $\beta = m - d/2 > 0$. Then the Sobolev space $H^m([0,1]^d)$ is a bounded-kernel RKHS and*
$$C^m([0,1]^d) \hookrightarrow H^m([0,1]^d) \hookrightarrow \Lambda^\beta([0,1]^d).$$
The same conclusion holds with the larger classical Lipschitz endpoint $C^{m-1,1}([0,1]^d)$ in place of $C^m([0,1]^d)$.

Theorem 2 (Purely classical Holder convention)**.** *Let $d \geq 1$, $\beta > 0$, and $\alpha = \beta + d/2$. Interpret C^s classically: $C^s = C^{\lfloor s \rfloor, s - \lfloor s \rfloor}$ for noninteger s, and C^s is the usual space of s-times continuously differentiable functions when $s \in \mathbb{N}$. Then there exists an RKHS H with*
$$C^\alpha([0,1]^d) \hookrightarrow H \hookrightarrow C^\beta([0,1]^d)$$
if and only if $\alpha \in \mathbb{N}$ and $\beta \notin \mathbb{N}$. Equivalently, the positive classical cases are exactly the odd-dimensional critical endpoints with integer domain smoothness.

More precisely, if $\alpha = m \in \mathbb{N}$ and $\beta = m - d/2 \notin \mathbb{N}$, then
$$C^{m-1,1}([0,1]^d) \hookrightarrow H^m([0,1]^d) \hookrightarrow C^\beta([0,1]^d).$$
If $\alpha = m \in \mathbb{N}$ and $\beta = \ell \in \mathbb{N}_0$, then no RKHS H satisfies
$$C^m([0,1]^d) \hookrightarrow H \hookrightarrow C^\ell([0,1]^d),$$
and the same negative conclusion holds with $C^{m-1,1}$ in place of C^m.

Thus Theorem 1 does not contradict the familiar one-dimensional positive endpoint
$$C^1([0,1]) \hookrightarrow H^1([0,1]) \hookrightarrow C^{1/2}([0,1]).$$
That positive result uses the smaller classical integer domain; the full Zygmund domain $\Lambda^1 = B^1_{\infty,\infty}$ is too large.

Idea of the Proof

The negative theorem is a multiscale cotype obstruction. At dyadic level j, place 2^{jd} separated smooth bumps of height $2^{-j\alpha}$. In the target Λ^β, each bump has norm comparable from below to $2^{-j(\alpha-\beta)}$. Since $\alpha-\beta=d/2$, one whole dyadic level contributes a fixed positive amount to the square-sum appearing in cotype 2.

The point that closes the integer endpoint is to estimate the domain norm by finite differences. If $r>\alpha$, then the r-th finite difference of the level-j part is bounded by

$$C\min\{2^{-j\alpha},\ |h|^r 2^{j(r-\alpha)}\}.$$

Summing this estimate over all levels gives $O(|h|^\alpha)$, uniformly in all signs and all finite sets of bumps. Hence the Rademacher norm in Λ^α is uniformly bounded, while the Λ^β square-sum over the first m levels grows like $\sqrt{m}$. This violates cotype 2, and the source's 2-factorability theorem excludes every intermediate RKHS.

The positive classical-integer statement is the standard Sobolev route: C^m and $C^{m-1,1}$ embed into $H^m=W^{m,2}$, while the critical Besov embedding sends $H^m=B^m_{2,2}$ into $B^{m-d/2}_{\infty,\infty}=\Lambda^{m-d/2}$. If $m-d/2$ is noninteger, this target is the classical Holder space $C^{m-d/2}$.

The remaining classical integer-target obstruction is Fourier-analytic. If the target is C^ℓ, then ℓ-th derivative evaluations are bounded on H. Applied to characters $e_n(x)=\exp(2\pi i n\cdot x)$, those derivatives contribute a factor of order $|n|^\ell$. The domain C^m, or even $C^{m-1,1}$, gives only $\|e_n\|\lesssim |n|^m$. At the critical gap $m-\ell=d/2$, Bessel's inequality would force the divergent series $\sum_{n\neq 0}|n|^{-d}$ to be bounded.

Finite Difference Norms

Let $r\in\mathbb{N}$. For a function f on $[0,1]^d$, define

$$\Delta_h^r f(x)=\sum_{\nu=0}^{r}(-1)^{r-\nu}\binom{r}{\nu}f(x+\nu h),$$

where only admissible x,h for which all points $x,x+h,\dots,x+rh$ lie in $[0,1]^d$ are used. If $r>s>0$, the Holder-Zygmund norm is equivalent to

$$\|f\|_{\Lambda^s}\simeq\|f\|_\infty+\sup_{0<|h|\le 1}|h|^{-s}\left\|\Delta_h^r f\right\|_\infty .$$

This is the standard finite-difference characterization of $B^s_{\infty,\infty}$ on cubes; see, for example, Triebel's function-space references [2].

Multiscale Bump Lemmas

Fix a nonzero function $\phi\in C_c^\infty((0,1/4)^d)$. For $j\ge 0$ and $k=(k_1,\dots,k_d)\in\{0,\dots,2^j-1\}^d$, define

$$\phi_{j,k}(x)=2^{-j\alpha}\phi(2^j x-k),\qquad x\in[0,1]^d.$$

For each fixed j, the supports of the functions $\phi_{j,k}$ are separated, and there are 2^{jd} of them.

Lemma 1 (Uniform signed domain bound)**.** *For every $\alpha > 0$ there is a constant C_α such that for every finite set*

$$A \subset \{(j,k) : j \geq 0,\ k \in \{0,\dots,2^j - 1\}^d\}$$

and every choice of signs $\varepsilon_{j,k} \in \{-1,1\}$,

$$\left\| \sum_{(j,k)\in A} \varepsilon_{j,k}\phi_{j,k} \right\|_{\Lambda^\alpha([0,1]^d)} \leq C_\alpha.$$

Proof. For each level j, set

$$g_j = \sum_{k:(j,k)\in A} \varepsilon_{j,k}\phi_{j,k}.$$

Because the supports at a fixed level are separated, for every integer $r > \alpha$,

$$\|g_j\|_\infty \leq C2^{-j\alpha}, \qquad \|D^r g_j\|_\infty \leq C_r 2^{j(r-\alpha)}.$$

Thus for every admissible h,

$$\|\Delta_h^r g_j\|_\infty \leq C_r \min\{2^{-j\alpha},\ |h|^r\, 2^{j(r-\alpha)}\}.$$

The first bound is the trivial finite-difference bound by the sup norm, and the second follows from the integral representation of the r-th finite difference in terms of $D^r g_j$.

Let $0 < |h| \leq 1$, and choose $J \geq 0$ with

$$2^{-(J+1)} < |h| \leq 2^{-J},$$

with the evident modification for $|h| > 1/2$. Then

$$\begin{aligned}\sum_{j\geq 0} \|\Delta_h^r g_j\|_\infty &\leq C_r\,|h|^r \sum_{j\leq J} 2^{j(r-\alpha)} + C_r \sum_{j>J} 2^{-j\alpha} \\ &\leq C_{\alpha,r}\,|h|^\alpha\,.\end{aligned}$$

The sup norm is also uniformly bounded:

$$\left\| \sum_j g_j \right\|_\infty \leq C \sum_{j\geq 0} 2^{-j\alpha} < \infty.$$

The finite-difference norm equivalence proves the lemma. □

Lemma 2 (Target lower bound)**.** *For every $\beta > 0$ there is $c_\beta > 0$ such that, for all j,k,*

$$\|\phi_{j,k}\|_{\Lambda^\beta([0,1]^d)} \geq c_\beta\, 2^{-j(\alpha-\beta)}.$$

Proof. Choose an integer $r > \beta$. Since ϕ is nonzero and compactly supported, its r-th finite differences are not identically zero. Hence there exist $x_0 \in (0,1/4)^d$, $h_0 \in \mathbb{R}^d$, and $\eta > 0$ such that the points $x_0, x_0 + h_0, \dots, x_0 + rh_0$ remain in $(0,1/4)^d$ and

$$\left|\Delta_{h_0}^r \phi(x_0)\right| \geq \eta.$$

Put $x_{j,k} = 2^{-j}(k + x_0)$ and $h_j = 2^{-j}h_0$. Scaling gives

$$\Delta_{h_j}^r \phi_{j,k}(x_{j,k}) = 2^{-j\alpha}\Delta_{h_0}^r \phi(x_0).$$

Therefore

$$\|\phi_{j,k}\|_{\Lambda^\beta} \geq c\,|h_j|^{-\beta} \left|\Delta_{h_j}^r \phi_{j,k}(x_{j,k})\right| \geq c_\beta 2^{-j(\alpha-\beta)}.$$

□

Proof of Theorem 1

Let

$$A_m = \{(j,k) : 0 \le j < m,\ k \in \{0,\dots,2^j-1\}^d\}.$$

By Lemma 1, for independent Rademacher signs $\varepsilon_{j,k}$,

$$\mathbb{E}\left\| \sum_{(j,k)\in A_m} \varepsilon_{j,k}\phi_{j,k} \right\|_{\Lambda^\alpha} \le C_\alpha.$$

On the other hand, Lemma 2 and the critical identity $\alpha - \beta = d/2$ give

$$\begin{aligned}\left(\sum_{(j,k)\in A_m} \|\phi_{j,k}\|_{\Lambda^\beta}^2 \right)^{1/2} &\ge c_\beta \left(\sum_{j=0}^{m-1} 2^{jd} 2^{-2j(\alpha-\beta)} \right)^{1/2} \\ &= c_\beta \sqrt{m}.\end{aligned}$$

Thus the cotype-2 constants of the inclusion

$$\Lambda^\alpha([0,1]^d) \hookrightarrow \Lambda^\beta([0,1]^d)$$

are unbounded. The inclusion is not cotype 2.

If an RKHS H satisfied

$$\Lambda^\alpha([0,1]^d) \hookrightarrow H \hookrightarrow \Lambda^\beta([0,1]^d),$$

then Schölpple and Steinwart's 2-factorization theorem would make the inclusion $\Lambda^\alpha \hookrightarrow \Lambda^\beta$ 2-factorable [1, Theorem 6]. Their type/cotype obstruction says every 2-factorable inclusion is cotype 2 [1, Lemma 5 and Corollary 8]. This contradiction proves the theorem.

Proof of Proposition 1

Assume $m \in \mathbb{N}$ and $d < 2m$. Then $m > d/2$, so the Sobolev space

$$H^m([0,1]^d) = W^{m,2}([0,1]^d)$$

embeds continuously into $C^0([0,1]^d)$. Point evaluations are therefore bounded, and $H^m([0,1]^d)$ is a bounded-kernel RKHS on the compact cube.

The inclusion

$$C^m([0,1]^d) \hookrightarrow H^m([0,1]^d)$$

is immediate because all weak derivatives up to order m are bounded on the cube. The same is true for $C^{m-1,1}$, since its derivatives of order m exist weakly and belong to L^∞.

Finally, the standard critical Besov embedding gives

$$H^m([0,1]^d) = B^m_{2,2}([0,1]^d) \hookrightarrow B^{m-d/2}_{\infty,2}([0,1]^d) \hookrightarrow B^{m-d/2}_{\infty,\infty}([0,1]^d) = \Lambda^\beta([0,1]^d),$$

where $\beta = m - d/2 > 0$. This proves the asserted RKHS sandwich. $\square$

Derivative Character Obstruction

The next lemma handles the classical integer target C^ℓ. It is a derivative version of the finite-character argument used for the bounded-kernel endpoint.

Lemma 3. *Let $m, \ell \in \mathbb{N}_0$, and let E be a Banach space of functions on $[0,1]^d$ containing the characters*

$$e_n(x) = \exp(2\pi i n \cdot x), \qquad n \in \mathbb{Z}^d.$$

Assume that for some $M > 0$,

$$\|e_n\|_E \le M(1 + |n|)^m, \qquad n \in \mathbb{Z}^d.$$

If the inclusion $E \hookrightarrow C^\ell([0,1]^d)$ factors through a Hilbert space, then

$$\sum_{n \in \mathbb{Z}^d \setminus \{0\}} \frac{|n|^{2\ell}}{(1 + |n|)^{2m}}$$

has uniformly bounded finite partial sums. In particular, if $m - \ell = d/2$, no such Hilbert factorization exists.

Proof. Suppose there are a Hilbert space $\mathcal{H}$ and bounded linear maps

$$J : E \to \mathcal{H}, \qquad R : \mathcal{H} \to C^\ell([0,1]^d)$$

such that $RJf = f$ for all $f \in E$. Put $a = \|J\|$ and $b = \|R\|$. Real spaces may be complexified, so it suffices to argue over complex scalars. We work on the torus through the standard half-open section $\sigma : \mathbb{T}^d \to [0,1]^d$.

For every multi-index γ with $|\gamma| = \ell$ and every $t \in \mathbb{T}^d$, the functional

$$h \longmapsto D^\gamma Rh(\sigma(t))$$

has norm at most b. Let $v_t^\gamma \in \mathcal{H}$ be its Riesz representative, so that

$$D^\gamma Rh(\sigma(t)) = \langle h, v_t^\gamma \rangle_{\mathcal{H}}, \qquad \|v_t^\gamma\| \le b.$$

Fix a finite $F \subset \mathbb{Z}^d \setminus \{0\}$, and let

$$E_F = \operatorname{span}\{Je_n : n \in F\} \subset \mathcal{H}$$

with orthogonal projection P_F. For $n \in F$, define

$$u_n^\gamma = \int_{\mathbb{T}^d} e_n(t)\, P_F v_t^\gamma \, dt.$$

Vector-valued Parseval/Bessel gives, for each fixed γ,

$$\sum_{n \in F} \|u_n^\gamma\|_{\mathcal{H}}^2 \le b^2.$$

Since $Je_n \in E_F$,

$$\begin{aligned} \langle Je_n, u_n^\gamma \rangle &= \int_{\mathbb{T}^d} D^\gamma RJe_n(\sigma(t))\, \overline{e_n(t)}\, dt \\ &= (2\pi i n)^\gamma. \end{aligned}$$

Therefore

$$\|u_n^\gamma\| \geq \frac{(2\pi)^\ell\, |n^\gamma|}{aM(1+|n|)^m}.$$

Consequently, for every γ with $|\gamma| = \ell$,

$$\sum_{n\in F} \frac{|n^\gamma|^2}{(1+|n|)^{2m}} \leq C_\gamma a^2 b^2 M^2.$$

Summing over all $|\gamma| = \ell$, and using

$$\sum_{|\gamma|=\ell} |n^\gamma|^2 \geq c_{d,\ell}\, |n|^{2\ell}$$

with the evident interpretation when $\ell = 0$, yields the asserted uniform finite-partial-sum bound.

If $m - \ell = d/2$, the lower estimate

$$\frac{|n|^{2\ell}}{(1+|n|)^{2m}} \gtrsim |n|^{-d}$$

on dyadic annuli contradicts the divergence of $\sum_{n\in\mathbb{Z}^d\setminus\{0\}} |n|^{-d}$. □

Proof of Theorem 2

If $\alpha \notin \mathbb{N}$, then $C^\alpha = \Lambda^\alpha$. Since $C^\beta \hookrightarrow \Lambda^\beta$ for every $\beta > 0$, any RKHS sandwich

$$C^\alpha \hookrightarrow H \hookrightarrow C^\beta$$

would also give

$$\Lambda^\alpha \hookrightarrow H \hookrightarrow \Lambda^\beta,$$

contradicting Theorem 1. Hence no positive classical case exists with noninteger α.

Now let $\alpha = m \in \mathbb{N}$. If $\beta = m - d/2 \notin \mathbb{N}$, then $\Lambda^\beta = C^\beta$, and Proposition 1 gives the positive sandwich

$$C^{m-1,1} \hookrightarrow H^m \hookrightarrow C^\beta.$$

In particular, the same holds with the smaller domain C^m.

It remains to exclude the case $\beta = \ell \in \mathbb{N}_0$. Then

$$m - \ell = d/2.$$

For both $E = C^m([0,1]^d)$ and $E = C^{m-1,1}([0,1]^d)$, the characters satisfy

$$\|e_n\|_E \leq M(1+|n|)^m.$$

Lemma 3 therefore rules out every Hilbert factorization of $E \hookrightarrow C^\ell([0,1]^d)$, and in particular rules out every intermediate RKHS. This proves the classification.

Consequences and Scope

The critical line is now fully settled for the Holder-Zygmund scale $\Lambda^s = B^s_{\infty,\infty}$: at

$$\alpha - \beta = d/2, \qquad \beta > 0,$$

there is no intermediate RKHS. This includes integer α, closing the gap left by the earlier noninteger-only multiscale argument.

At integer smoothness, however, "Holder" has several common meanings. The Zygmund endpoint $B^m_{\infty,\infty}$ is larger than C^m and $C^{m-1,1}$. Theorem 2 gives the complete classical convention map: the purely classical critical sandwich is positive exactly when the domain exponent is an integer but the target exponent is not an integer. Equivalently, this positive case occurs in odd dimensions with integer domain smoothness. When both endpoint exponents are integers, the derivative-character obstruction gives a negative answer.

Together with the companion one-dimensional metric Holder packet, this explains the apparent endpoint split: the classical $C^1([0,1]) \to H^1([0,1]) \to C^{1/2}([0,1])$ sandwich is real, but it does not extend to the larger Zygmund domain $\Lambda^1([0,1])$.

Verification and Novelty Check

The proof is noncomputational. The main verification points are:

1. the finite-difference estimate in Lemma 1, which is the step that handles integer α;
2. the target lower bound in Lemma 2;
3. the critical count $2^{jd}2^{-2j(\alpha-\beta)} = 1$;
4. the source paper's implication from RKHS sandwich to 2-factorability and hence cotype 2;
5. the standard Sobolev/Besov embedding used in Proposition 1;
6. the derivative-character obstruction in Lemma 3, which rules out the classical integer-target case.

The run indexes were searched for arXiv id 2312.14711, Holder-Zygmund, Besov $B_{\infty,\infty}$, $\alpha-\beta = d/2$, integer alpha, cotype, and 2-factorability. Existing local hits were the bounded-kernel endpoint full packet, the one-dimensional companion packet, and the noninteger-alpha partial packet now subsumed by Theorem 1.

Bounded web searches on June 16, 2026 for the source title plus "Holder-Zygmund border", for "$B^s_{\infty,\infty}$ cotype Hilbert factorization", for "Besov $B_{\infty,\infty}$ 2-factorable", and for "Holder-Zygmund RKHS critical $d/2$" found the source arXiv record but no exact later answer to this critical Holder-Zygmund line. Additional searches for "C^m Hilbert factorization C^ℓ critical Sobolev embedding", "C^m reproducing kernel Hilbert C^ℓ critical", and "Holder critical RKHS C^m C^ℓ" found no exact classical endpoint convention classification.

Human Review Recommendation

Review as a likely valid full result for the Holder-Zygmund/Besov critical line and for the purely classical endpoint convention. The key checks are that the finite-difference proof of Lemma 1 really gives a uniform Λ^α bound for arbitrary signs and arbitrary finite collections of dyadic bumps, and that Lemma 3 correctly converts C^ℓ-target derivative evaluations into the critical lattice-sum contradiction.

References

[1] Max Schölpple and Ingo Steinwart. *Which Spaces can be Embedded in Reproducing Kernel Hilbert Spaces?* arXiv:2312.14711, 2023. `https://arxiv.org/abs/2312.14711`.

[2] Hans Triebel. *Theory of Function Spaces.* Birkhäuser, 1983.

[3] Thomas Runst and Winfried Sickel. *Sobolev Spaces of Fractional Order, Nemytskij Operators, and Nonlinear Partial Differential Equations.* De Gruyter, 1996.